GAME THEORY & NASH EQUILIBRIA — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Game Theory & Nash Equilibria

Question 1 10.0 points

Examine the extensive form game tree in the diagram below for a sequential two-player game where Player 1 can choose between actions A and B, and Player 2 can then choose between X and Y after observing Player 1's action. Identify the subgame perfect Nash equilibrium by performing backward induction on this game tree.

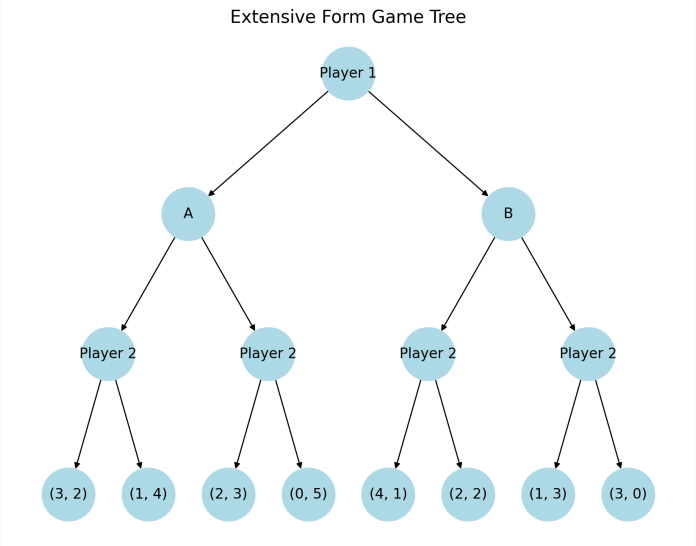

Fig. 1. An extensive form game tree for a sequential two-player game where Player 1 chooses between actions A and B, and Player 2 chooses between X and Y after observing Player 1's action. The tree includes decision nodes, actions, and payoffs.

Question 2

10.0 points

Consider the payoff matrix for a Prisoner's Dilemma shown in the diagram below: | | Cooperate | Defect | |-------|----------|--------| | C | (3,3) | (0,5) | | D | (5,0) | (1,1) | Identify all Nash equilibria in this game and explain why they are Nash equilibria.

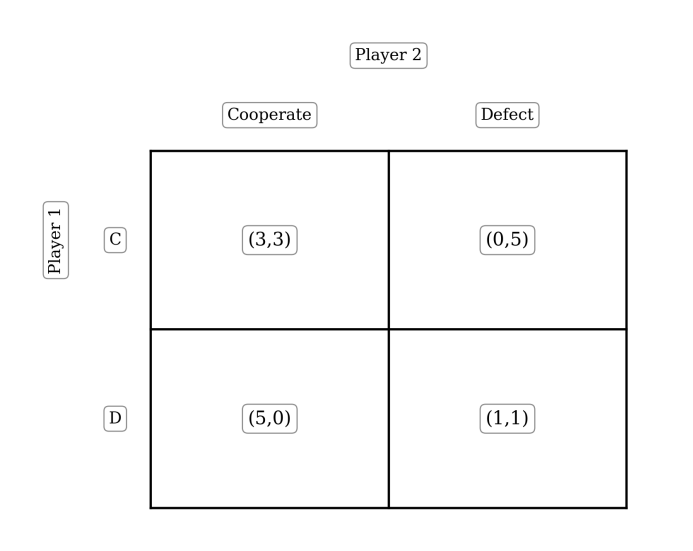

Fig. 2. A payoff matrix for a Prisoner's Dilemma game. The matrix has two players with strategies 'Cooperate' and 'Defect'. The payoffs are: (3,3) for both cooperating, (0,5) for player 1 cooperating and player 2 defecting, (5,0) for player 1 defecting and player 2 cooperating, and (1,1) for both defecting. The matrix should clearly label the strategies and payoffs for each player.

Question 3 10.0 points

Evaluate a mixed strategy Nash equilibrium in a coordination game given by the payoff table in the diagram below, where both players choose high or low effort: | | High | Low | |----------|-------|-------| | High | (2,2) | (0,3) | | Low | (3,0) | (1,1) | Use the mixed strategy method to find the equilibria, and determine the probabilities with which each player should randomize their strategies.

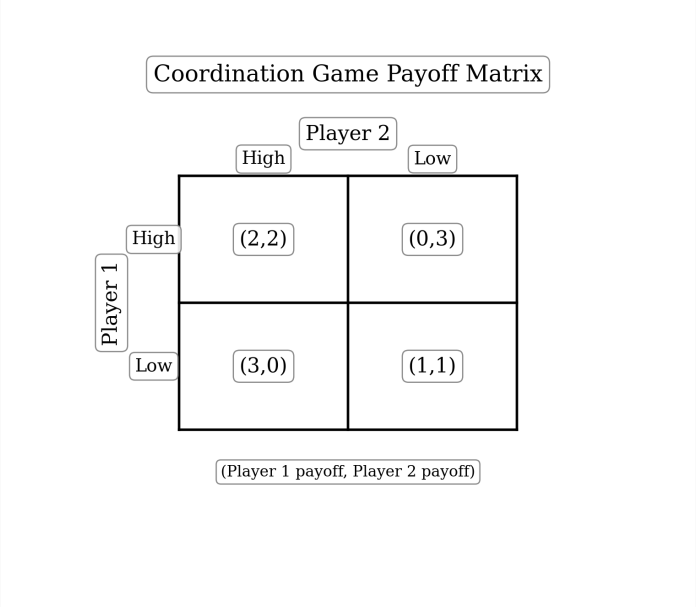

Fig. 3. A payoff matrix for a coordination game where both players choose high or low effort. The matrix shows payoffs for each combination: (High, High) = (2,2), (High, Low) = (0,3), (Low, High) = (3,0), (Low, Low) = (1,1).

Question 4

10.0 points

Using the diagram below, perform backward induction on the game tree for a three-stage sequential game with perfect information. Each player has one possible optimal move at each decision node. Determine the optimal strategy for each player, and explain how backward induction aids in determining this strategy.

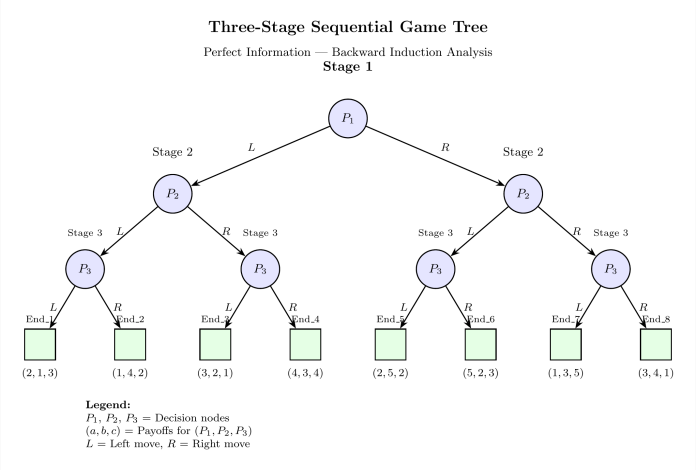

Fig. 4. A three-stage sequential game tree with perfect information. The tree includes decision nodes for each player and possible moves at each node, illustrating the sequence of play. Each player has one possible optimal move at each decision node. IMPORTANT: The diagram is missing required labels. Specifically: The diagram lacks payoff values (e.g., a pair of numbers like (3, 5)) at each of the terminal nodes (End_1. The diagram is missing the payoff values at the terminal nodes, which are absolutely essential for performing backward induction as requested by the question.. Add each missing label as visible text at the appropriate location in the diagram.

Question 5 10.0 points

Examine the Pareto frontier in the diagram below, which illustrates efficient versus inefficient outcomes. Identify the Pareto efficient points and describe the trade-offs at inefficient points. Explain how any shift in the curve can alter efficiency in allocation.

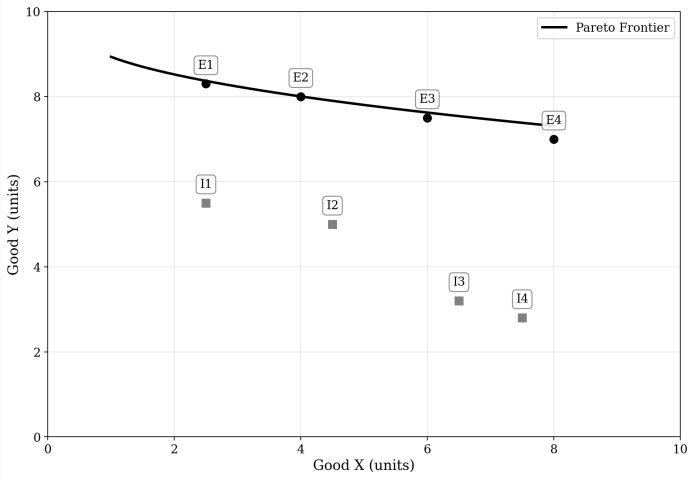

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): The diagram contains a legend that explicitly labels the curve as 'Pareto Frontier', which directly answers the part of the question asking the student to identify the efficient outcomes.; The diagram explicitly labels the curve as the "Pareto Frontier" in the legend, which directly answers the question asking the student to identify the Pareto efficient points.. Remove the legend and its label 'Pareto Frontier'. The diagram should show a 2D plot with 'Good X (units)' on the x-axis and 'Good Y (units)' on the y-axis. Plot a downward-sloping curve. Mark several points on the curve (e.g., E1, E2, E3, E4) and several points below the curve (e.g., I1, I2, I3, I4, I5). Do not label the curve itself as the Pareto Frontier. IMPORTANT correction needed: The diagram explicitly labels the curve as the "Pareto Frontier" in the legend, which directly answers the question asking the student to identify the Pareto efficient points.. Specifically: The legend contains the label 'Pareto Frontier', which is the concept the student is supposed to identify. The DIAGRAM DESCRIPTION. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.